

Affan Muhammad Shaikh

Oshawa, ON | 647-309-1927 | ffaanshake@gmail.com | [linkedin.com/in/sil6428](https://www.linkedin.com/in/sil6428) | github.com/sil6428 | [Portfolio](#)

Education

Ontario Tech University Oshawa, ON
Bachelor of Information Technology - Networking & Cybersecurity Sept. 2024 - Expected 2028
– Relevant coursework: Advanced Networking I, Cybersecurity Foundations, Programming I with Python, and Computer Systems.

Certification

CompTIA Security+ | In progress Expected Sept. 2026

Cybersecurity Projects

CICIDS2017 Reproduction and Split-Sensitivity Study | Python, pandas, scikit-learn, pytest [GitHub](#)
– Audited 2,830,743 labeled network-flow records across 8 source files, removed 2,867 non-finite rows, verified duplicate-feature behavior, and preserved row-level provenance through deterministic pipelines.
– Reproduced a Random Forest baseline on 239,603 sampled flows across 15 labels, reaching 99.57% mean random-split accuracy over 3 seeds with 14 passing automated tests.
– Measured a 78.79-point accuracy gap between random splits and leave-one-source-file-out evaluation, plus 14.43% exact train-test feature-and-label overlap, exposing evaluation leakage and unseen-class limits.

Network Security Lab | Cisco IOS, Packet Tracer, Wireshark

- Configured IPv4/IPv6 addressing, subnetting, VLANs, access ports, 802.1Q trunks, inter-VLAN routing, DHCP, STP, and OSPF/EIGRP fundamentals in routed and switched topologies.
- Diagnosed connectivity and routing faults using ping, traceroute, interface state, routing tables, Cisco show commands, and packet captures.

File Integrity Monitor | Python, SHA-256, JSON, unittest

- Built a command-line monitor that creates trusted SHA-256 baselines and reports added, modified, deleted, and moved files through deterministic JSON evidence and script-friendly exit codes.
- Detected 45 of 45 controlled changes across 500 fixture files with zero scan errors, including same-size tampering and rename detection, while passing 7 automated tests.

Technical Skills

Security: IDS/IPS evaluation, dataset validation, vulnerability analysis, system hardening, authentication, authorization, RBAC, firewalls, encryption and hashing concepts, incident-response fundamentals
Networking: IPv4/IPv6, subnetting, VLANs, 802.1Q, inter-VLAN routing, DHCP, DNS, NAT, STP, OSPF/EIGRP fundamentals, routing and switching troubleshooting
Programming and Data: Python, pandas, NumPy, scikit-learn, pytest, TypeScript, JavaScript, React, Next.js, HTML, CSS
Systems and Tools: Linux, Windows Server, Cisco IOS, Wireshark, Packet Tracer, SecureCRT, Git, GitHub, Cloudflare Workers, Proxmox

Additional Technical Work

Interactive 3D Portfolio | TypeScript, React, Three.js, Cloudflare Workers [GitHub](#)
– Built and deployed an interactive 3D workspace with custom models, responsive camera controls, a simulated desktop environment, keyboard and touch support, and automated route checks.

Archtech Nonprofit Technology Operations | Google Workspace, Web Hosting, Team Coordination

- Set up Google Workspace for a developing nonprofit and coordinate the team currently building its full website.
- Own the website hosting and deployment workflow while contributing a smaller share of the implementation.

Experience and Leadership

Sales Associate May 2025 - Present
Winners Oshawa, ON
– Process customer transactions accurately, resolve product questions, and coordinate inventory and floor organization in a fast-paced retail environment.

Community Volunteer 400 Hours

Al Arqam Islamic Centre Oshawa, ON

- Supported registration, guest service, crowd flow, setup, cleanup, and attendee needs at large community events.